

Chandrabhas Nalabotu

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<https://101cn.github.io/portfolioCN/>

Education

University of Leicester - Mechanical Engineering MEng

2024 - Present

- Year 1 and 2 - 1st

Work experience / Projects

Aerolabs CFD Analyst / Avionics engineer - UAS Challenge

September 2025 - Present

- Ran 3D CFD simulations in STAR-CCM+ on a drone airframe, increasing lift coefficient from 1.0 to 1.6 by adding secondary flaps, enabling lower minimum flight speed
- Reduced CFD simulation runtime from 6 hours to ~2 hours by tuning solver settings and convergence criteria under senior team member guidance
- Configured flight controller optimising control delivery and sensitivity

Innex1 Rover - Lunabotics 2026

September 2025 - Present

- Built a 5-topic ROS2 architecture with dedicated bridges for drivetrain, excavation and actuator subsystems, reducing controller input to rover response latency from 2.5s to 0.6s and enabling controller teleoperation via Pygame
- Reduced live camera feed latency from 30s to ~1s at 40Hz through bandwidth reduction over a ROS2/WebSocket pipeline to mission control
- Designed fail-closed motion safety logic with a 200ms stale-command timeout and stall-detection encoder verification to prevent uncontrolled spinning under low traction
- Separated 60A motor power and 15A compute power domains with independent fusing and a fault-latching E-stop circuit, preserving telemetry and compute uptime during an E-stop event

Single axis stabilizer - NEA coursework

September 2023 - May 2024

- Engineered a single-axis camera mount that actively counteracts external disturbance using IMU feedback while enabling continuous 360° rotation, controlled via Arduino
- Reduced angular tracking error by 30% and eliminated dropped interrupts by refactoring open-source stabilization control code
- Designed mechanical housing and PCB layout using Fusion 360 and EasyEDA, standardizing hole and via sizing to meet manufacturing tolerances

Technical skills:

Software - Solidworks, Fusion 360, Excel, STAR CCM+

Coding / Frameworks - Python, Arduino, C++, MATLAB, Simulink, ROS2, Linux distributions

Manufacturing - 3d prints, lathe, laser cutter, Soldering and standard workshop machining equipment